



2025

P24

BOOKLET NO.

108081

Forest Services

Mathematics

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.



SECTION – A

Q1. Solve **any five** out of seven.

(8×5=40)

Q1. (a) Derive the relationship between Cartesian (x, y) and Polar coordinates (r, θ) in 2D.

Convert the Cartesian equation $(x^2 + y^2)^2 = 4xy$ to polar form.

8

Q1. (b) Prove that a continuous function on a compact set is bounded.

8

Q1. (c) State and prove the necessary conditions for $f(z)$ to be analytic (Cauchy-Riemann Partial Differential Equations).

8

Q1. (d) Explain the concept of a 'Gradient' of a scalar field. Find the unit normal vector to the surface $x^2y + 2xz^2 - 8 = 0$.

8

Q1. (e) Prove that a sequence $\langle a_n \rangle$ of real numbers converges, if and only if it is a Cauchy sequence.

8

Q1. (f) Using Lagrange's interpolation formula find the value of y when $x = 10$ from the following data :

8

x	5	6	9	11
y	12	13	14	16



- Q1.** (g) A particle is projected with velocity u at an elevation α . Prove that if $h < \frac{u^2 \sin^2 \alpha}{2g}$, then the particle will be again at a height h after time t given by $\frac{2}{g} \sqrt{u^2 \sin^2 \alpha - 2gh}$ after being at the same height the first time. 8

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- Q2.** (a) Find the magnitude and the equations of the line of shortest distance between the lines

$$\frac{x-8}{3} = \frac{y+9}{-16} = \frac{z-10}{7} \text{ and } \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}. \quad 15$$

- Q2.** (b) State and prove the first Isomorphism theorem for groups. 15

- Q2.** (c) Let $\phi(x, y, z)$ and $\phi(x + \Delta x, y + \Delta y, z + \Delta z)$ be the temperature at two neighbouring points $P(x, y, z)$ and $Q(x + \Delta x, y + \Delta y, z + \Delta z)$ of a certain region. 10

i) Interpret physically the quantity $\frac{\Delta \phi}{\Delta s} = \frac{\phi(x + \Delta x, y + \Delta y, z + \Delta z) - \phi(x, y, z)}{\Delta s}$, where

Δs is the distance between points P and Q .

ii) Evaluate $\lim_{\Delta s \rightarrow 0} \frac{\Delta \phi}{\Delta s} = \frac{d\phi}{ds}$ and interpret physically.

iii) Show that $\frac{d\phi}{ds} = \nabla \phi \cdot \frac{d\vec{r}}{ds}$.



- Q3. (a)** Find a bilinear transformation which transform the unit circle $|z| = 1$ into real axis in such a way that the points $1, i, -i$ are mapped into $0, 1, \infty$ respectively. Into what regions the interior and exterior of the circle are mapped ? **15**

- Q3. (b)** Define divergence and curl of a vector field. Prove that $r^n \vec{r}$ is irrotational. Also find 'n' when it is solenoidal. **15**

- Q3. (c)** Define Jacobian for three variables. If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$, then find the Jacobian $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)}$. **10**

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- Q4. (a)** Reduce the equation $2x^2 + 7y^2 - 10yz - 8zx - 10xy + 6x + 12y - 6z + 2 = 0$ is an equation referred to principal axes as co-ordinate axes. **15**

- Q4. (b)** Given the values **15**

x :	5	7	11	13	17	—
f(x) :	150	392	1452	2366	5202	—

Evaluate $f(g)$ using Lagrange's and Newton's divided difference formulae.

- Q4. (c)** Prove $\nabla^2 r^n = n(n+1) r^{n-2}$ where n is a constant and also prove $\nabla^2 f(r) = \frac{d^2 f}{dr^2} + \frac{2}{r} \frac{df}{dr}$. **10**
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SECTION – B

Q5. Solve **any five** out of seven.

(8×5=40)

Q5. (a) Find the rank of given matrix A, where $A = \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$. 8

Q5. (b) Solve $\frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 - x^2)} = \frac{dz}{z(x^2 - y^2)}$. 8

Q5. (c) Evaluate integral $\int_0^6 \frac{dx}{1+x^2}$ by Simpson's $\frac{1}{3}$ rule. 8

Q5. (d) Explain bitwise operations AND, OR with examples. 8

Q5. (e) Determine the characteristic roots and the corresponding characteristic vectors of the

matrix A, where $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ 8

Q5. (f) Find the complete integral of $p^2x - q^2y = z$ by using Charpit method. 8

Q5. (g) Evaluate the integral $\int_0^1 \frac{dx}{1+x^2}$ by Trapezoidal rule. 8



Q6. (a) Solve $\frac{d^2y}{dx^2} - y = xe^{-x}$. **15**

Q6. (b) Find the orthogonal trajectories of the family of curves $r^n = a^n \cos n\theta$. **15**

Q6. (c) Convert the **10**

i) Binary number 101010 to decimal number.

ii) Octal number $(25)_8$ to decimal number.

Q7. (a) Using Runge – Kutta method of fourth order, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2, 0.4$. **15**

Q7. (b) Solve the following transportation problem by using Vogel's approximation method. **15**

	D₁	D₂	D₃	Supply
O₁	2	7	4	5
O₂	3	3	1	8
O₃	5	4	7	7
O₄	1	6	2	14
Demand	7	9	18	34

Q7. (c) State and prove D'Alembert principle. **10**



- Q8. (a) Solve $3x^4p^2 - xp - y = 0$. 15
- Q8. (b) Construct the Lagrangian for a particle moving in a space and then deduce the equation of motion. (Use Cartesian coordinates) 15
- Q8. (c) Solve the following minimal assignment problem. 10

Man → Job ↓	a	b	c	d
I	12	30	21	15
II	18	33	09	31
III	44	25	24	21
IV	23	30	28	14

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